

Case 2: ALFWorld Heat Task (Egg → Microwave → Countertop)

Task: Heat some egg and put it in countertop.

Core Skills	[hea_001] Secure exact target first; [hea_003] Open-Place-Heat sequence; [hea_004] No appliance before object.
Step 1-3	Obs: Countertop 1 (no egg) → Countertop 2 (egg found). Action: take egg 1 Reasoning: [hea_004] Avoid microwave until object is held. [hea_002] Systematic search of surfaces.
Step 4-6	Obs: Microwave 1 is closed. Action: open microwave 1 → heat egg 1 Reasoning: [hea_003] Correct sequence: open door first, then initiate state change.
Step 7-8	Obs: Egg is heated. Action: go to countertop 1 → put egg 1 Reasoning: [hea_005] Direct Post-Heat Placement: move to target without detours.

Result: SUCCESS (8 Steps) — Skills used: [hea_001] (Target First), [hea_004] (Sequence Control).